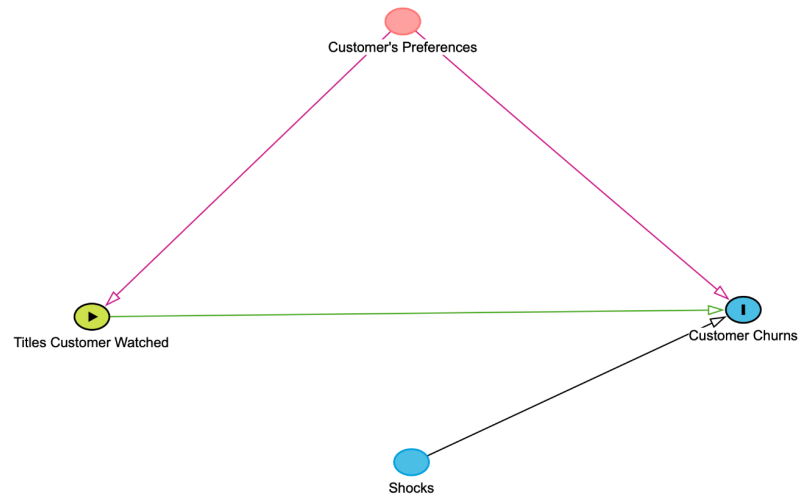


Content Optimization

The goal of this project is to assign titles to reduce their content catalogue while minimizing churn and maximizing their revenue. The identification problem is assigning how much each title contributes to churn in their catalogue. This is hard to quantify due to the various confounding factors that influence a customer to churn as well as watching particular titles.

The DAG model problem illustrates this problem. We are looking to quantify the titles a customer watches and how those titles contribute to retention. Now there are other factors that will contribute to churn such as seasonality, and price shocks. These will be included the best it can be in order to wipe away this variation. Now customer preferences will affect both churn and the titles they watch. BritBox is a British television service so this in turn tells us something about customer preferences already, that they enjoy British programs. Customer's preferences can affect both which titles they watch and the likelihood they churn. If a customer likes lifestyle genres this will influence them to mostly watch lifestyle shows and therefore if BritBox drops a majority of their lifestyle titles this will increase the likelihood this customer will churn. The key issue here is quantifying an individual title's effectiveness in retention.



The key methodology is shown below. The majority of the issue is identifying change in probability or Δp_{ij} where X_i is a given title. We would want to measure the change in probability of churning when removing a given title. This is where the issue of causal identification comes in, in exactly how we quantify this delta. However, after this delta is calculated it is aggregated to obtain a retention metric in dollars for each title. V_{i_ret} is the retention metric which gets the expected revenue a title is contributing to by looking at the expected change in probability for each customer times the current price they are paying. This would then be inputted into the objective function taking into account the cost of each title in order to maximize revenue given the budget and genre diversity constraint. This will tell us what titles we should then keep in the catalogue.

$$\Delta p_{ij} = P(\text{retain} \mid x_i) - P(\text{retain} \mid \text{w/o } x_i)$$

$$V_i^{\text{ret}} = \sum_{j=1}^{NM} MP \cdot \Delta p_{ij} + \sum_{j=1}^{NA} AP \cdot \Delta p_{ij}$$

NM : total monthly subscribers

NA : total annual subscribers

MP : monthly price

AP : annual price

$$\max_i (V_i^{\text{ret}} - c_i) x_i$$

Budget constraint:

$$\sum_i c_i x_i \leq B_{\max}$$

Genre diversity constraint:

$$\sum_{i \in G_n} x_i \geq m_g$$

For initial ranking use a churn model and calculate the delta, Probability of Retention (with title) - Probability of Retention (without title) by setting watch time = 0 in a given month. Using the churn model this will predict the probability of churning given the customer did not watch the title, but is not a causal takeaway. This provides preliminary ranking and sensitivity of a title according to the churn model. A key limitation of this model is that there is no substitution effect where a customer does not move their time to another title. Another limitation is that these deltas can be looked at more so as correlation than causation because a customer can watch a title right before they churn however that may not be the cause. Because of this, this title may show a negative delta where removing that title actually increases retention when in reality it does not. Distinguishing these situations will be difficult however I do have a few methods to pursue to overcome this through propensity score matching via IPW and AIPW as well as

choice models. I am looking at using IPW and/or AIPW in order to compare customers who watched the show to customers who were just as likely to watch the show but didn't and compare their churn outcomes. A key assumption of this method is that at least 1 of the 2 models in AIPW needs to be correctly identified, either the outcome model or the probability model. Another assumption is that there is enough overlap with each title for those who were likely to watch compared to those who weren't. Another avenue would be looking at choice models by simulating what they would have watched if they didn't watch that show.

The robustness checks I plan to do are assessing the calibration of the model. Given the high data imbalance of 97% of the observations being subscribed compared to only 3% churning it will be very difficult to have a well calibrated model. Will aim to use loss functions and evaluation metrics such as MCC in order to train this unbalanced model effectively. Will be performing time split training on the previous timeframe and predicting on the new timeframe. There will be some titles that were actually dropped in this test set which I will then use my predictions to compare what actually occurred.